The magnet

MRI safety physics for the anaesthetist, written for the paediatric list.

What the ten cards ask you to do on the next MRI list, one line from each.

1 Always on

Treat the magnet room as live at every hour of every day (MHRA; Wilson).

2 The pull

Check everyone and everything at the door, every time: the child, the parents, the staff, the trolley, the monitor and the pumps (Wilson; MHRA).

3 The twist

Ask about metal in the eye, old injuries and shrapnel, since the eye and tissue near vessels are at particular risk from a fragment that turns (Wilson).

4 The line

Ask where the 0.9 mT and 0.5 mT lines run in your own unit, and whether either reaches the control room or the corridor (MHRA).

5 Labels

Read the numbers on the label, and compare them with the field and the gradient at the place the kit will stand (MHRA).

6 Heat

Straighten every cable, never run one diagonally across the child, pad between the skin and any lead, and pad between the arms and the body so no loop of skin closes; a blanket is not padding (ASA; MHRA).

7 Noise

Fit MR Safe ear plugs, ear defenders or both to the child before the scan, and record it in the notes (Wilson; MHRA).

8 The ECG in the bore

Expect the ST segment and T wave to change as the table goes in, and use the ECG in the bore for rate and rhythm (Wilson; ASA).

9 The arrest

Agree before the list who stops the scan, who calls, who opens the doors and where the resuscitation area is; Wilson asks for it in a written procedure.

10 The quench

Know your unit's quench procedure before your first list there, and where the quench buttons are, in the magnet room and the control room, with the time the field takes to fall written beside them (Wilson; MHRA).

Dr Ganesh Sivasankara · MD · FRCA · FCARCSI · Consultant Anaesthetist

The cards, with the pictures and the sources, at **gundoc9.github.io/the-magnet/**

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